Introduction

- **U.S. decarbonization efforts are underway in the power sector, but industry and transportation remain difficult to decarbonize with limited options…**
    - U.S. decarbonization will require efforts across all sectors, with shares of greenhouse gas emissions by economic sector being transportation (28%), power (25%), industry (23%), residential & commercial (13%), and agriculture (10%) (*EPA 2026*).
    - Of the top 3 emitting sectors, decarbonization efforts within the power sector are commercially available and well underway, with 51% of planned utility-scale capacity additions in 2026 being solar and 14% being wind (*EIA 2026*).
    - For the transportation and industrial sector, the decarbonization transition is more difficult. Options involve direct electrification – for example of vehicles themselves or for electrified steam crackers, retrofits with carbon capture, and fuel switching (H2 fuel cell vehicles, or switching from fossil fuels to bio and synthetic fuels).
- **With limited options to decarbonize transportation, electrification of the LDV sector is a key strategy that has already been commercially deployed. Due to the Renewable Fuel Standard, this would signal ethanol demand decline in the U.S. which has economic and political implications.**
    - Ethanol in the U.S. primarily comes from 191 corn ethanol plants which produce a total of 18,477 MMgal/year (*EIA 2025*). Ethanol production is tied to gasoline demand at around 10% of gasoline by volume because of the blend wall, which cannot be exceeded without further retrofits to car infrastructure to prevent corrosion.
    - Reacting to this disruption in the corn ethanol industry and establishing a long term plan is crucial because of how pervasive corn ethanol, and the associated corn production, is in the Midwest economy and political interests.
- **Policymakers have a few options to respond to anticipated ethanol demand decline….**
    - **Ethanol demand decline could be an opportunity to retire the corn ethanol industry**, and thus around 40% of corn that supplies the corn ethanol feedstock. This is a position well advocated for by environmental advocates who cite the environmental burdens of continued corn usage, and would also require a clear transition plan for alternative uses for the corn land.
    - **Policymakers could also seek to maintain the corn ethanol industry** and thus related corn land to preserve existing job infrastructure, and instead explore alternative end uses for corn ethanol beyond blending into gasoline, such as upgrading to liquid fuels and also dehydrating into ethylene. [ethylene has high emission/energy intensity – what is the case for ethylene from ethanol? Why not other chemicals] Using corn ethanol as a feedstock has the potential to help decarbonize the liquid fuels and ethylene sectors. There is already a demonstrated interest in this path from the Biden administration via the 45Z Clean Fuel Production tax credit, which was lowered but still exists after OBBBA, and investment in the commercial viability of alcohol-to-jet plants such as LanzaJet.
    - Another reason why ethanol to ethylene is because landfilling ethylene at end of life could be a form of carbon sequestration, and with a bio-source particularly with CCS could potentially have a net negative emissions credit.
- **Evaluating these two options requires consideration of the environmental, economic, and societal impacts of choosing to retire corn and corn ethanol or not. We seek to provide guidance for corn ethanol's potential role in solely the energy system as a starting point.** We ask – what is the cost-optimal disposition of existing corn ethanol assets in a decarbonized system? How might this change in different net zero future scenarios with various industry decisions, market factors, and resource uncertainties?

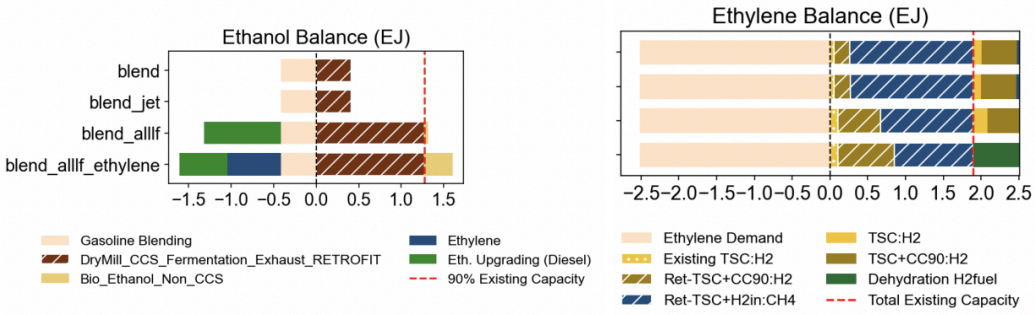
- **Previous energy systems models that parameterize corn ethanol** include a range of representations of corn ethanol production with carbon capture retrofits. Literature about decarbonizing corn ethanol production itself via carbon capture and fuel switching is primarily evaluated via TEA or via a process-level design. Previous energy systems modelling studies involve a limited range of potential end uses for corn ethanol, primarily focused on ethanol blending into gasoline and/or ethanol upgrading solely to jet fuel based on current commercial and policy interest.
    - Deep decarbonization and broad level retrofits and technology options such as electrified steam crackers, high levels of CO2 capture from ethanol plants, not captured in previous studies
- **The following study, using a multi-sector energy systems capacity expansion model** seeks to elevate the diverse range of options explored via TEA and process-design to both produce corn ethanol with a lower carbon intensity and to upgrade it to liquid fuels and ethylene. Evaluating these options for corn ethanol production and usage within a system-level framework is necessary because of key system wide linkages – limited resources such as existing corn ethanol production, energy crop biomass, and CO2 storage availability as well as cross-sector linkages such as increased electricity demand via implementation of carbon capture retrofits.
    - We co-optimize corn ethanol production with retrofits and alternative technologies for liquid fuel and ethylene production (we both optimize ethanol production and link it to ethanol upgrade, and we also put them against alternative technologies such as retrofits to steam crackers and bio-jet fuel production)

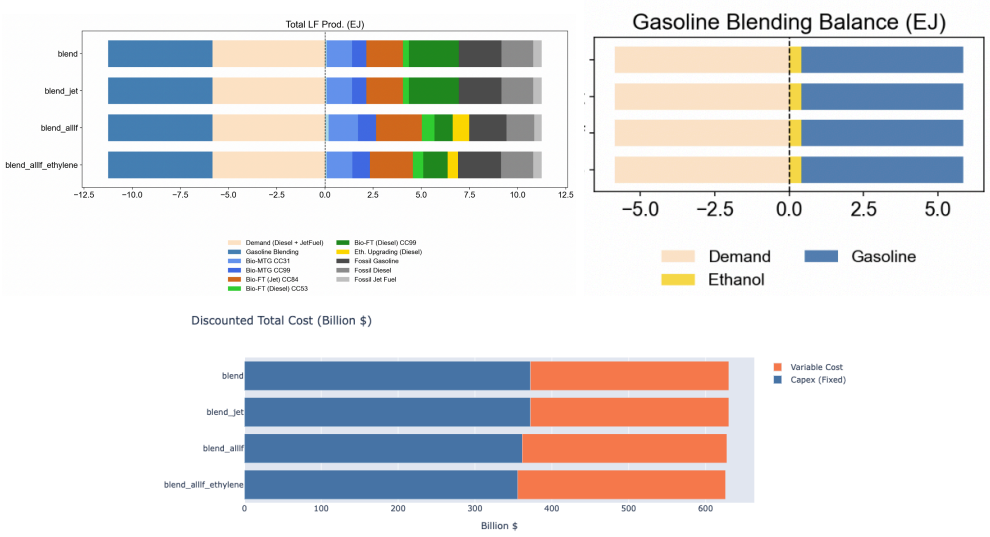
Results

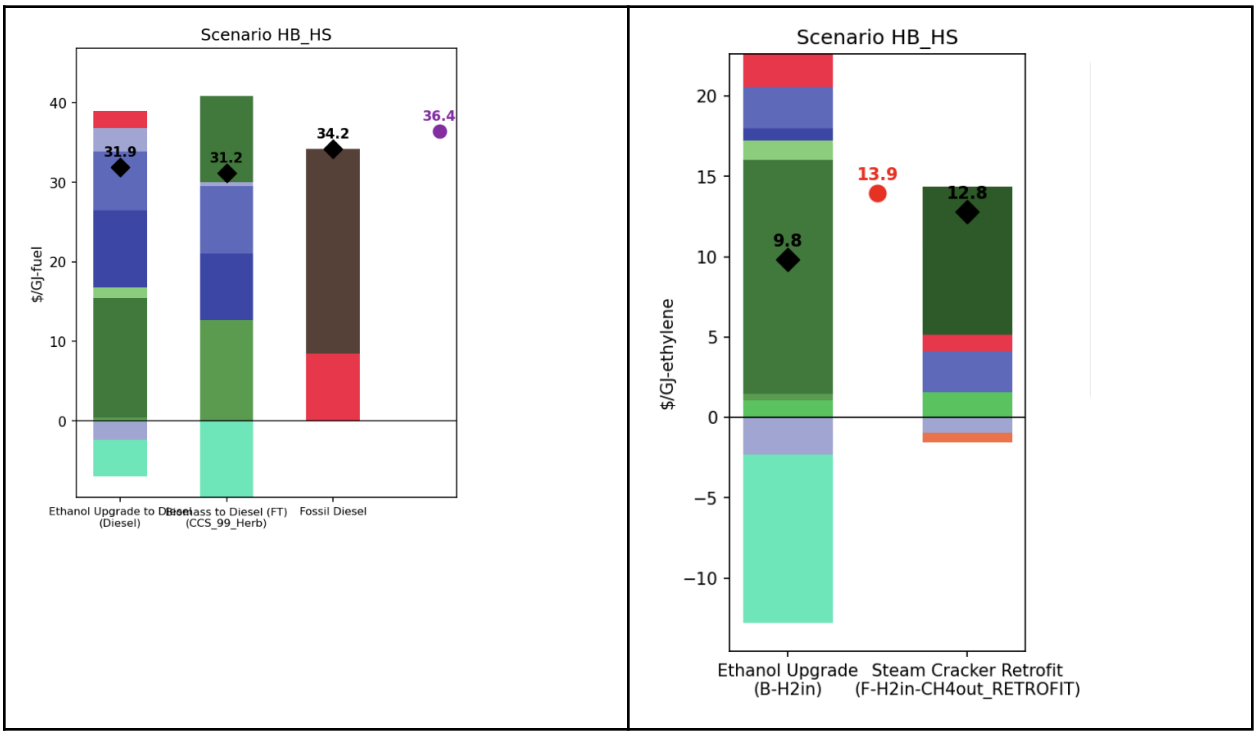
Increasing corn ethanol end use options beyond ethanol-to-jet decreases system-wide cost. For the resource case of high biomass and high CO2 storage availability, we test a range of scenarios with increasing end use options for corn ethanol usage. When corn ethanol has the maximum end use optionality, with the ability to blend into gasoline, upgrade to liquid fuels, or dehydrate to ethylene, the cost-optimal solution shows a blend of all three end pathways.

We find that corn ethanol can be valuable in a decarbonized system as a strategy to help decarbonize the liquid fuels and ethylene sectors.

1. Within the *liquid fuels sector*, corn ethanol feedstock value is its potential to abate the use of fossil diesel, which in this case study is more expensive than fossil jet fuel or fossil gasoline
2. Within the *ethylene sector*, corn ethanol feedstock value is that it provides a lower CI value than ethane-based steam cracker retrofit ethylene production alternatives.







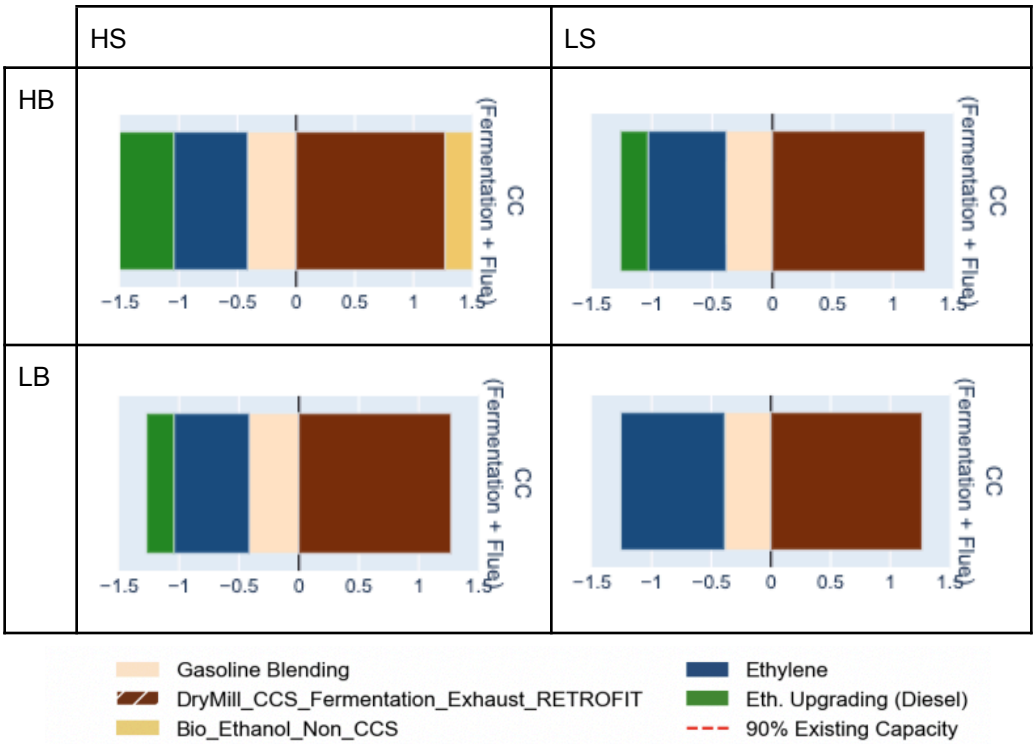
Cost-optimal corn ethanol end usage varies based on biomass and CO2 storage resources scenarios, and the ethanol dehydration to ethylene pathway demonstrates a more structural advantage than ethanol upgrading to liquid fuels due to carbon abatement benefits. Previously showed just one iteration of a possible resource future for energy crop biomass and subsurface CO2 storage availability – how might the potential uses of corn ethanol differ if we test different resource future scenarios?

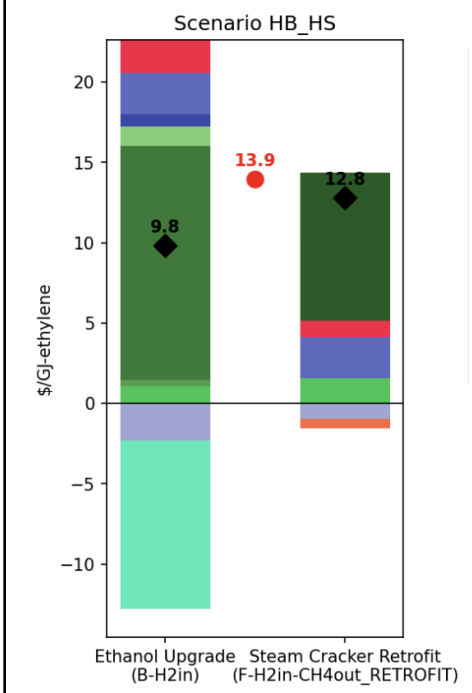
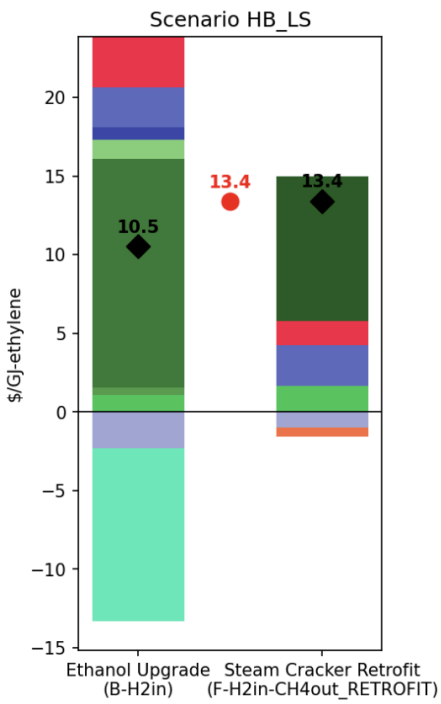
The question of ethylene vs liquid fuels is, what is the value of capturing and storing the carbon? – and for

the LS case, we calculate the modified capture cost as the credit you get from the emissions abated, but you add the cost of ADDING to the storage pool, HOWEVER when we use ethylene as a permanent CO2 storage, we are conveniently AVOIDING the CO2 storage. So then it's just about, at what CO2 price does the abatement value of going to diesel or ethylene become comparable?

We can just clearly see that ethylene not only has abatement value, but is has value in the system as a CO2 negative process

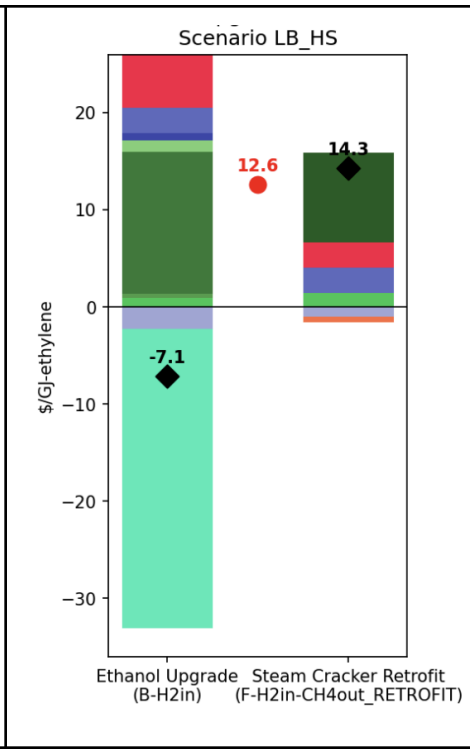
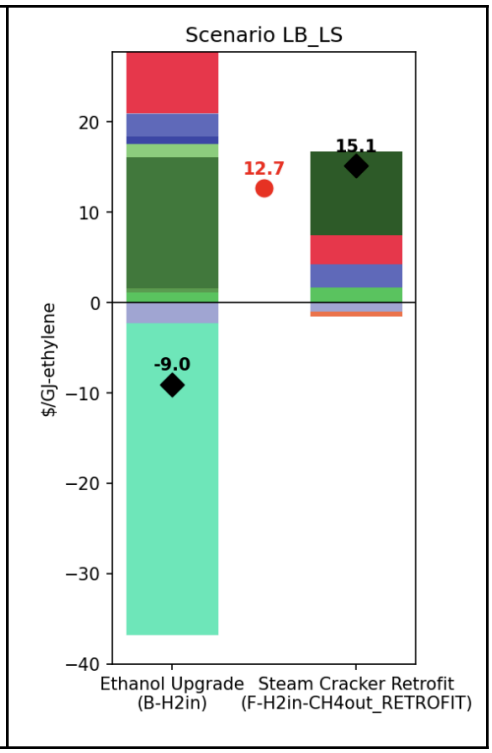
1. With higher CO2 capture credits, whether it's because of low CO2 storage availability or lack of other CO2 capture options with low biomass availability, corn ethanol will route to ethylene because of the greater CI difference between a bio-based and fossil-based ethylene pathway – liquid fuels sector has other bio-based alternatives with gasification and FT – plan is to justify this with the carbon abatement cost plot
2. We find that in limited resource futures, either for biomass or for CO2 storage, that lead to a high carbon capture credit, the case for corn ethanol to be dehydrated into ethylene increases due to a higher value of corn ethanol to ethylene as a carbon sequestration method.

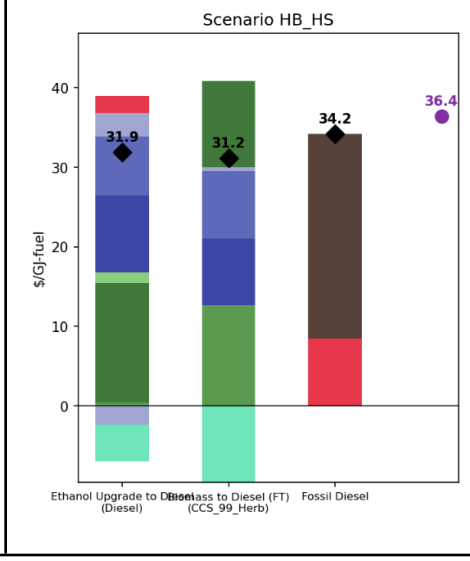
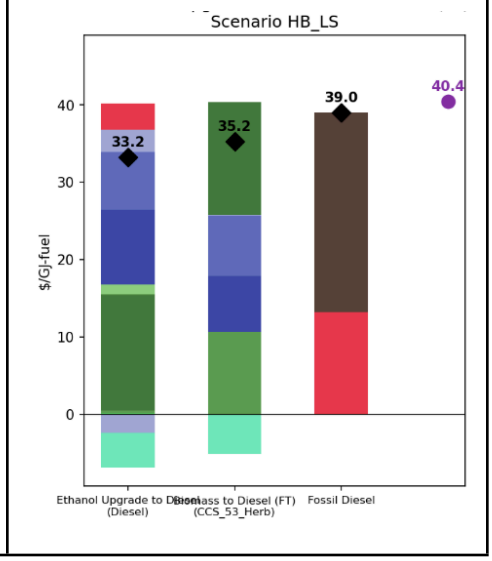


	HS	LS
HB		

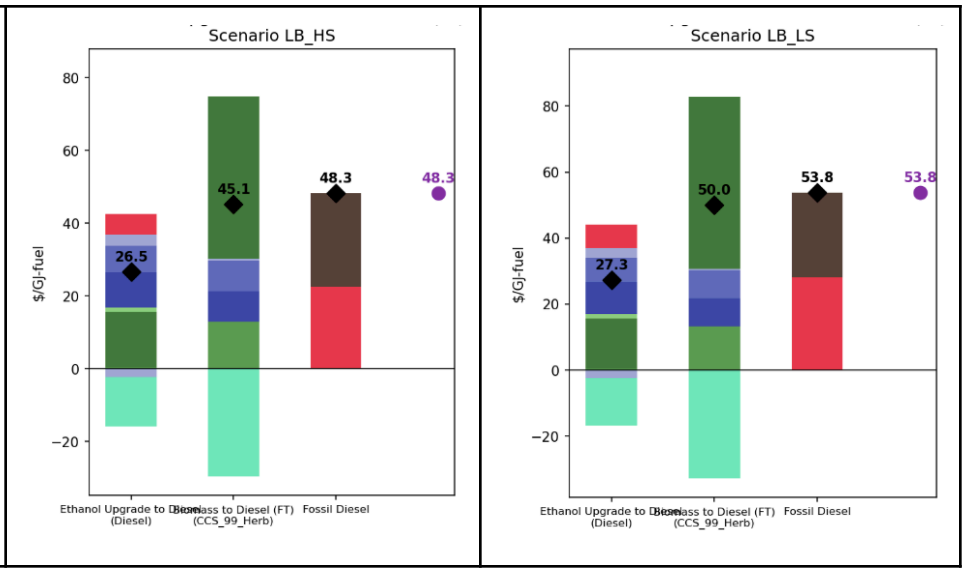
Scenario HB_HS
$/GJ-ethylene
20
15
10
5
0
−5
−10
13.9
9.8
12.8
Ethanol Upgrade
(B-H2in)
Steam Cracker Retrofit
(F-H2in-CH4out_RETROFIT)

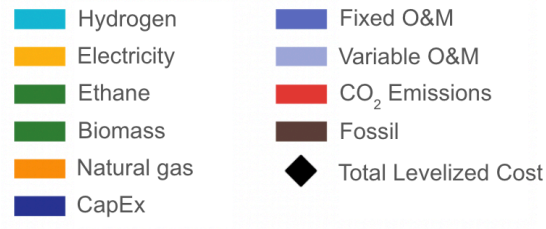
Scenario HB_LS
$/GJ-ethylene
20
15
10
5
0
−5
−10
−15
13.4
10.5
13.4
Ethanol Upgrade
(B-H2in)
Steam Cracker Retrofit
(F-H2in-CH4out_RETROFIT)

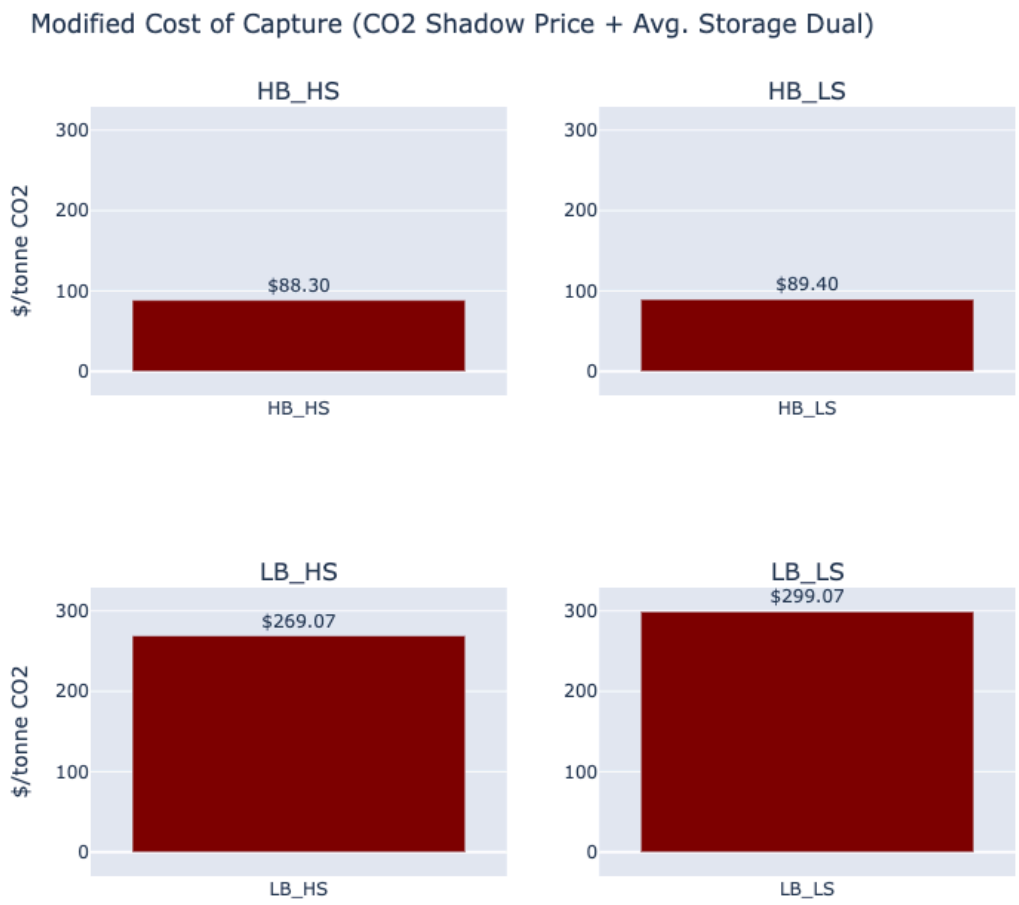
LB	 Scenario LB_HS $/GJ-ethylene 12.6 14.3 -7.1 Ethanol Upgrade (B-H2in) Steam Cracker Retrofit (F-H2in-CH4out_RETROFIT)	 Scenario LB_LS $/GJ-ethylene 12.7 15.1 -9.0 Ethanol Upgrade (B-H2in) Steam Cracker Retrofit (F-H2in-CH4out_RETROFIT)

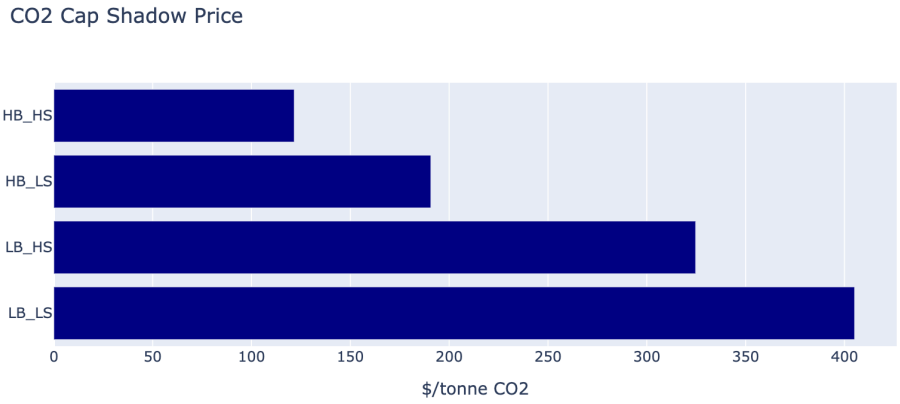
	HS	LS
HB	 Scenario HB_HS $/GJ-fuel 31.9 31.2 34.2 36.4 Ethanol Upgrade to Diesel (Diesel) Biomass to Diesel (FT) (CCS_99_Herb) Fossil Diesel	 Scenario HB_LS $/GJ-fuel 33.2 35.2 39.0 40.4 Ethanol Upgrade to Diesel (Diesel) Biomass to Diesel (FT) (CCS_53_Herb) Fossil Diesel

LB


Scenario LB_HS
Scenario LB_LS
$/GJ-fuel
80
60
40
20
0
−20
26.5
45.1
48.3
48.3
27.3
50.0
53.8
53.8
Ethanol Upgrade to Diesel (Diesel)
Biomass to Diesel (FT) (CCS_99_Herb)
Fossil Diesel


Hydrogen
Electricity
Ethane
Biomass
Natural gas
CapEx
Fixed O&M
Variable O&M
CO_2 Emissions
Fossil
Total Levelized Cost





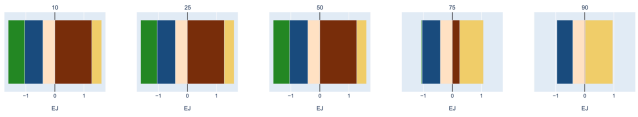
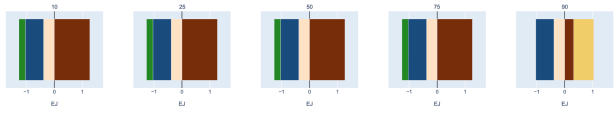
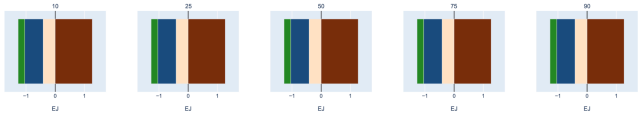
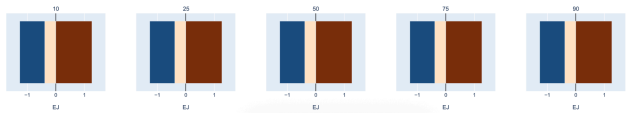
Could show a CAC plot here with similar magnitudes….value in both sectors…value is pretty much comparable

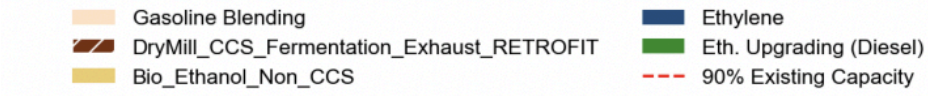
Since we previously described how the corn ethanol to ethylene pathway is strengthened by limited carbon capture options, we test how sensitive this is to our policy assumption that the CO2 in ethylene is sequestered at end of life. [I have the results for this test, but have not yet made the plots]

Sensitivity: Test how sensitive ethanol to liquid fuel allocation is to fossil liquid fuel costs. Currently, we use fossil liquid fuel costs of $70.56/MWh-jet fuel, $92.81/MWh-diesel, $69.57/MWh-gasoline, but we test a case where we flip the jet fuel and diesel price and predict that ethanol will be used in the sector where fossil price is the highest. The choice to upgrade ethanol to X liquid fuel would just depend on predictions about those fossil fuel prices for the future.

Sensitivity: we investigate how various corn prices may change the cost-optimal disposition of corn ethanol in the decarbonized system. Corn prices can change due to a range of market factors, and we also acknowledge that our baseline corn price at $160/t-corn is an exogenous assumption, since there are other corn demand categories that are not captured in this energy systems model. We test a corn price sweep based on a historical data set: 10th percentile = $134/t-corn, 25th percentile = $142/t-corn, 50th percentile = $160/t-corn, 75th percentile = $203/t-corn, 90th percentile = $246/t-corn.

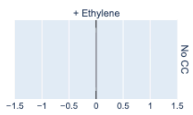
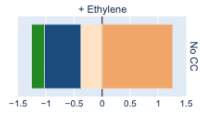
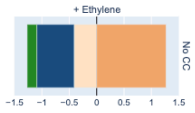
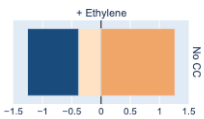
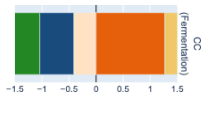
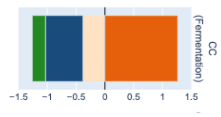
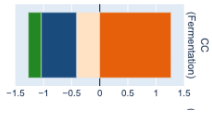
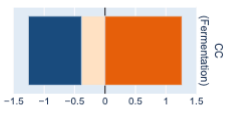
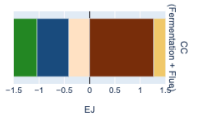
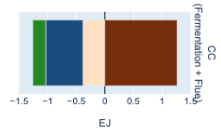
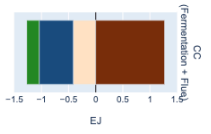
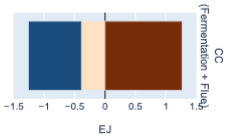
- Ethanol itself is still upgraded to both liquid fuels and ethylene, however in a high biomass case, facing higher corn prices, corn ethanol itself is replaced with new build cellulosic ethanol plants with lower feedstock costs.
- In the 75th and 90th percentile cases, we see retirements in corn ethanol and then replacements

	HS	LS
HB		
LB		



Sensitivity: we investigate the path dependency on ability to retrofit with CC and the usage of corn ethanol in the system.

- We see that a similar amount of corn ethanol is deployed even despite no CC options – shows that there is a tolerance in the range of CI for corn ethanol that is still economic and still benefits from its MUCH lower CI than for the fossil-ethylene scenarios
- What is the added CO2 network capacity in the last row, and how might that not be feasible – so we explore the options where there is limited CO2 infrastructure buildout that the ethanol industry can use – maybe because of competition or just not being able to build it out – compare the buildout between the scenarios

No CC				
CC fermentation				
CC fermentation + flue gas				



*lighter orange is no CC and darker orange is fermentation CC only

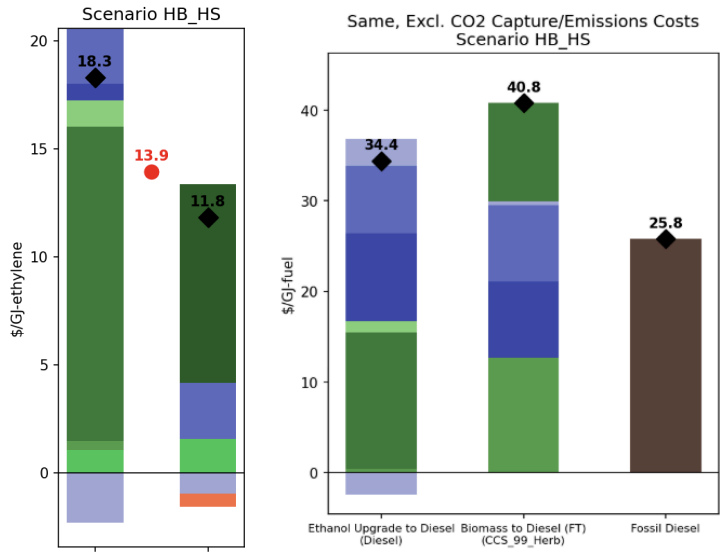
Discussion and conclusions

Expanding corn ethanol's end-use options beyond ethanol-to-jet fuel, which is currently supported by policy and industry, offers system-wide cost savings.

Our model relies on high degrees of sequestration from ethanol plants concentrated in the Midwest, but what kind of CO2 infrastructure from the Midwest to the US gulf coast is needed to enable that? Potentially, if it's going to be too not feasible to build this CO2 infrastructure, then perhaps we can just not do it.

What is holding us back from choosing bio-ethylene? What is the cost difference RIGHT NOW without a carbon price between conventional ethylene and also bio-ethylene?

- What is the current green premium gap between bioethylene and also regular? – is it that there is LESS of a gap between them right now (without a carbon price) so that's why we have been pushing ethanol towards jet fuel, for example?



Across market scenarios, upgrading ethanol to ethylene consistently holds value: ethanol dehydration is a compelling bio-based alternative to steam-cracker pathways that rely on ethane feedstock. However, under conditions that raise corn prices relative to lower-cost energy crop biomass, new cellulosic ethanol plants become economically viable. This suggests the ethanol-upgrading industry has lasting value, even with the possibility of corn ethanol plant retirement as new cellulosic ethanol plants fill that role.

